Phase I: Business Process Modeling

Project Title: VoltGuard Rwanda: Electricity Billing and Meter Tamper Monitor

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Course: INSY 8311 - Database Development with PL/SQL

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1. Define the Business Process

- **Selected Process Title:** Meter Reading and Billing Process
- **Purpose:** To capture monthly household electricity consumption, validate the reading against historical records, apply the correct versioned Rwanda Utilities Regulatory Authority (RURA) tariff, and generate an accurate end-user electricity bill.
- **Objectives:**
    - Record and normalize raw monthly electricity meter consumption data.
    - Validate incoming meter readings to detect and reject regressive or faulty values.
    - Calculate energy charges accurately using the tariff version valid on the specific reading date.
    - Generate and store customer bills while logging all verification steps for audit purposes.
- **The Problem it Addresses:** Utilities frequently suffer from non-technical losses—such as billing errors or unrecorded consumption caused by faulty meters, outdated tariff rates, or incorrect manual entries. For instance, failing to apply the October 2025 RURA tariff update or accepting a meter reading lower than the previous month results in lost revenue and billing disputes.
- **Importance to the Organization:** Implementing a structured, automated validation and billing workflow ensures financial accuracy, compliance with regulatory tariff structures, transparency for customers, and the prevention of revenue leakage for the utility provider.

- **Non-Technical Losses:** Electricity that is consumed or lost due to administrative errors, billing mistakes, meter tampering, or unmetered supply rather than physical power line loss.
- **Regressive Reading:** A new meter reading that is lower than the previous month's recorded reading (which is physically impossible under normal operation and indicates an error or tampering).
- **Tariff Bands:** Consumption tiers (such as 0–20 kWh) where different pricing rates apply based on how much electricity a household uses.
- **Audit Trail:** A chronological record of system actions and data changes that allows staff to track who changed a reading or generated a bill and when

2. Actors and Responsibilities

Actor	Type	Responsibility
Customer	User / Employee	Consumes electricity and checks/looks up their monthly electricity bill through the ORDS interface.
Field Meter Reader / Staff	User / Employee	Collects or inputs monthly meter readings into the system using the Oracle APEX application, and handles billing inquiries.
System Admin	User / Employee	Manages system access, enters new RURA tariff versions and bands into the database when updated, and monitors system logs.
RURA (Rwanda Utilities Regulatory Authority)	External Stakeholder	Regulates the utility sector and officially publishes updated end-user electricity tariffs. (External entity; provides the tariff rules applied by staff).
Database & Billing System (Oracle / PL/SQL)	Other Systems	Automatically validates meter readings (checking for regressive/lower values), looks up the active tariff version based on the reading date, calculates the bill, and maintains audit logs.

4: Data-Generating Activities (CRUD Table)

Step	Actor	Operation (CRUD)	Records / Tables Touched	Information Generated / Used

1. Manage Tariffs	System Admin	**Create / Update**	`TARIFF_VERSION` & `TARIFF_BAND`	Enters new RURA tariff structures and consumption block limits; updates the end date of previous tariff versions when a new one takes effect.
2. Submit Reading	Field Meter Reader / Staff	**Read**	`METER`	Looks up customer meter details and inputs the current meter reading value, date, and meter ID.
3. Validate & Check Tampering	Database & Billing System	**Create**	`TAMPER_EVENT`	If the new reading is lower than the previous one, the system rejects it and logs the attempted value, previous value, and timestamp.
4. Record Valid Reading	Database & Billing System	**Create**	`METER_READING`	Stores the validated meter reading, consumption date, and reading value.
5. Calculate & Bill	Database & Billing System	**Read / Create**	`TARIFF_VERSION`, `TARIFF_BAND`, & `BILL`	Looks up active tariff rates to compute consumption costs and generates the final customer bill record.
6. Review Tamper Flags	Management Team	**Update**	`TAMPER_EVENT`	Reviews flagged security incidents and updates their investigation or resolution status.
7. Track System Activity	Database & Billing System	**Create**	`AUDIT_LOG`	Automatically records transaction logs tracking data additions, changes, and administrative actions.
8. View District Reports	Management Team	**Read**	`METER_READING` & `CUSTOMER`	Retrieves monthly consumption summaries grouped by district for operational planning.

9. Customer Lookup	Customer	**Read**	`BILL` & `METER_READING`	Accesses the online portal to view generated monthly bills and billing history.
10. Archive / Purge Data	System Admin	**Delete / Archive**	`AUDIT_LOG` & `METER_READING`	*Note:* Historical records or old audit logs exceeding retention policy timelines are archived or purged to maintain database performance.